Week 14

MICROECONOMIC THEORY
ECON 323 502/503

Spring 2026

Associate Professor Huiyi Guo

Texas A&M University

Review

In sequential-move games, we adopt the solution concept of **Subgame Perfect Nash Equilibrium** (in this class), in which:

1. the follower plays its optimal strategy, i.e., takes the optimal action in response to every possible action taken by the leader;

2. anticipating that the follower will play the optimal strategy, the leader chooses its own optimal strategy.

Oligopoly Market

- An oligopoly market is a market with a few sellers. (different from perfect competitive market and monopoly market)
- We focus on the duopoly case: a market with two sellers.
- Each seller may have partial influence on the market's total output level or price.
- Sellers in oligopoly market can be viewed as players in a game.

Chapter 12. Monopolistic Competition and Oligopoly

Related Reading in Textbook

- 12.2 Oligopoly
- 12.3 Price Competition
- 12.4 Competition versus Collusion: The Prisoners' Dilemma

Models we study:

<u>Cournot model</u>

<u>Bertrand model</u>

<u>Stackelberg model</u>

Cournot model: Oligopoly model in which each firm treats the output of its competitors as fixed, and all firms decide simultaneously how many units to produce.

- **Strategy of each firm**: choosing an output level
- **Payoff of each firm**: profit level
- **Cournot Equilibrium**: Nash Equilibrium in the Cournot model.

How are firms' payoffs interconnected in a Cournot model?

- **Total output produced by firms** jointly decides the market price.

Example: Europe's Energy Crisis

In 2021, the European Union imported 83% of its natural gas.

Gas import sources

- Russia: 31.4%
- Norway: 22.7%
- LNG (liquified natural gas mainly from the US, Qatar and Nigeria): 25.7%
- Algeria: 10.7%, etc.

These sellers are oligopolists in the European natural gas market.

These countries' profits from natural gas sales are interconnected through the market price of natural gas.

Example: Europe's Energy Crisis

Since the start of the war in Ukraine, Europe has made efforts to reduce dependence on supply from Russia and the natural gas price has surged.

Example: Europe's Energy Crisis

The fallout from the war is making Norway richer.

Norway expects to bring in about $109 billion from the petroleum sector this year — $82 billion more than in 2021.

Sources:

- https://www.oecd.org/ukraine-hub/policy-responses/how-vulnerable-is-european-manufacturing-to-gas-supply-conditions-01278ba3/
- https://www.consilium.europa.eu/en/infographics/eu-gas-supply/
- https://www.washingtonpost.com/world/2022/10/08/norway-gas-prices-supply-europe/

Example (Review a Monopoly Problem). Suppose that Q(p) =1000 − 2p, there is a monopoly firm in the market with C(q) = 200q. What is the equilibrium output level, price level, and profit level in this market?

Solution:

Step 1: Derive $\boldsymbol{p(q), R(q), MR(q), MC(q)}$

Step 2: Set $\boldsymbol{MR(q^*) = MC(q^*)}$ and isolate $\boldsymbol{q^*}$**.** Compute $\boldsymbol{p(q^*)}$

Step 3 (Skip verifying no shut down condition): Derive profit

Example (Review a Monopoly Problem). Suppose that Q(p) =1000 − 2p, there is a monopoly firm in the market with C(q) = 200q. What is the equilibrium output level, price level, and profit level in this market?

Solution:

Step 1: Derive $\boldsymbol{p(q), R(q), MR(q), MC(q)}$

- $p(q) = 500 - 0.5q$
- $R(q) = p(q)q = (500 - 0.5q)q = 500q - 0.5q^2$
- $MR(q) = R'(q) = 500 - q$
- $MC(q) = C'(q) = 200$

Step 2: Set $\boldsymbol{MR(q^*) = MC(q^*)}$ and isolate $\boldsymbol{q^*}$**.** Compute $\boldsymbol{p(q^*)}$

- $500 - q^* = 200$. This leads to $q^* = 300$.
- $p(q^*) = 500 - 0.5q^* = 500 - 0.5 * 300 = 350$

Step 3 (Skip verifying no shut down condition): Derive profit

- $\Pi(q^*) = p(q^*)q^* - C(q^*) = 350 * 300 - 200 * 300 = 45000$

Example (Review a Competitive Market Problem). Suppose the market demand curve is given by Q(p) =1000 − 2p. Each firm in the competitive market has a cost function C(q) = 200q. What is the competitive equilibrium output level, price level, and profit level in this market?

Solution:

Step 1: Set $\boldsymbol{p = MC(q)}$. Derive the individual and market supply curves.

Step 2: Derive the competitive equilibrium.

Step 3: Derive profit

Example (Review a Competitive Market Problem). Suppose the market demand curve is given by Q(p) =1000 − 2p. Each firm in the competitive market has a cost function C(q) = 200q. What is the competitive equilibrium output level, price level, and profit level in this market?

Solution:

Step 1: Set $\boldsymbol{p = MC(q)}$. Derive the individual and market supply curves.

Individual supply curve is horizontal at $p = 200$. So is the market supply curve.

Step 2: Derive the competitive equilibrium.

Market supply curve: $\mathrm{p} = 200$

Market demand curve: Q(p)=1000-2p=600.

Competitive equilibrium price is 200 and quantity is 600.

Step 3: Derive profit

Each firm earns **zero** profit since p=AC.

Example (Cournot Equilibrium). Suppose that Q(p) =1000 − 2p, $C_1(q_1) = 200q_1$, and $C_2(q_2) = 200q_2$. What is the Cournot equilibrium output level, price level, and profit level in this market?

Solution:

Step 1(1): Derive the inverse demand function

Step 1(2): $\boldsymbol{R_1}$

Step 1(3) : Compute $\mathbf{MR_1}$ (take derivative of $\boldsymbol{R_1}$ with respect to q_1), $\mathbf{MC_1}$

Step 2: Set $\boldsymbol{MR_1 = MC_1}$

Example (Cournot Equilibrium). Suppose that Q(p) =1000 − 2p, $C_1(q_1) = 200q_1$, and $C_2(q_2) = 200q_2$. What is the Cournot equilibrium output level, price level, and profit level in this market?

Solution:

Step 1(1): Derive the inverse demand function

- $p(q) = 500 - 0.5q$, where $q = q_1 + q_2$.

Step 1(2): R_1

- $R_1 = p(q_1 + q_2)q_1 = (500 - 0.5q_1 - 0.5q_2)q_1 = 500q_1 - 0.5q_1^2 - 0.5q_1q_2$.

Step 1(3) : Compute MR_1 (take derivative of R_1 with respect to q_1), MC_1

- $MR_1 = 500 - q_1 - 0.5q_2$. $MC_1 = 200$.

Step 2: Set $MR_1 = MC_1$

- $500 - q_1 - 0.5q_2 = 200$, isolate q_1: $q_1 = 300 - 0.5q_2$ (1)
- **This is the Reaction curve of Firm 1, i.e. the optimal output level of Firm 1 given Firm 2's output level.**
- You can also write it as $q_1(q_2) = 300 - 0.5q_2$, i.e., highlight the fact that q_1 is a function of q_2

Example (Cournot Equilibrium). Suppose that Q(p) =1000 − 2p, $C_1(q_1) = 200q_1$, and $C_2(q_2) = 200q_2$. What is the Cournot equilibrium output level, price level, and profit level in this market?

Solution:

Step 1(1): Derive the inverse demand function

Step 1(2): $\boldsymbol{R_2}$

Step 1(3) : Compute $\mathbf{MR_2}$ (take derivative of $\boldsymbol{R_2}$ respect to q_2), $\mathbf{MC_2}$

Step 2: Set $\boldsymbol{MR_2 = MC_2}$

Example (Cournot Equilibrium). Suppose that Q(p) =1000 − 2p, $C_1(q_1) = 200q_1$, and $C_2(q_2) = 200q_2$. What is the Cournot equilibrium output level, price level, and profit level in this market?

Solution:

Step 1(1): Derive the inverse demand function

- $p(q) = 500 - 0.5q$, where $q = q_1 + q_2$.

Step 1(2): R_2

- $R_2 = p(q_1 + q_2)q_2 = (500 - 0.5q_1 - 0.5q_2)q_2 = 500q_2 - 0.5q_1q_2 - 0.5q_2^2$.

Step 1(3) : Compute MR_2 (take derivative of R_2 respect to q_2), MC_2

- $MR_2 = 500 - 0.5q_1 - q_2$. $MC_2 = 200$.

Step 2: Set $MR_2 = MC_2$

- $500 - 0.5q_1 - q_2 = 200$, isolate q_2: $q_2 = 300 - 0.5q_1$ (2)
- **This is the Reaction curve of Firm 2, i.e., the optimal output level of Firm 2 given Firm 1's output level.**
- You can also write it as $q_2(q_1) = 300 - 0.5q_1$, i.e., highlight the fact that q_2 is a function of q_1

Step 3. Solve the system of reaction curves to obtain the Cournot equilibrium quantity levels. Derive price.

$q_1 = 300 - 0.5q_2$ (1)

$q_2 = 300 - 0.5q_1$ (2)

Plug (1) into (2): $q_2 = 300 - 0.5(300 - 0.5q_2) = 300 - 150 + 0.25q_2 = 150 + 0.25q_2$.

Thus, $0.75q_2 = 150$. Hence, $q_2 = 150/0.75 = 200$.

Plugging it to (1), we have $q_1 = 300 - 0.5 * 200 = 200$.

Hence, $q_1 = q_2 = 200$. This is the Cournot equilibrium.

Equilibrium price: use the inverse demand function

p(q)=500-0.5q=500-0.5*(200+200)=300.

Step 4: Compute profit level of each firm.

Only Firm 1's output enters Firm 1's total cost function

Profits:

- $\Pi_1 = p(q_1 + q_2)q_1 - C_1(q_1) = p(q_1 + q_2)q_1 - 200\, q_1 = 300 * 200 - 200 * 200 = 20000.$

- $\Pi_2 = p(q_1 + q_2)q_2 - C_2(q_2) = p(q_1 + q_2)q_2 - 200\, q_2 = 300 * 200 - 200 * 200 = 20000.$

Only Firm 2's output enters Firm 2's cost function

Compare a Cournot duopoly model with a monopoly

Monopoly

- Output $q = 300$
- Price $p(q) = 350$
- Monopolist Firm's profit

$\Pi = 45000.$

Cournot Duopoly

- Output: $q_1 = q_2 = 200$
- Price $p(q_1 + q_2) = 300$
- Each firm's profit

$\Pi_1 = 20000.$
$\Pi_2 = 20000.$

Compare a Cournot duopoly model with a monopoly (summary)

Hence, a Cournot equilibrium has higher (total) output level, lower price, and lower (total) profit than a monopoly model.

unavailable in
noncooperative games

monopoly = perfect collude case

- When firms can perfectly collude via an enforceable contract, they will jointly agree on prices and quantities to produce.

- Firms can behave in aggregate like a monopolist (i.e., jointly produce the monopolist's profit maximizing output level or post the monopolist's profit maximizing price) and we call such an association a **cartel**.

- In the U.S., cartels are illegal.

- The Organization of Petroleum-Exporting Countries (OPEC); OPEC members jointly decide how much oil each member is allowed to produce.

Compare a Cournot duopoly model with competitive equilibrium

Market demand: Q(p) =1000 – 2p → $p(q) = 500 - 0.5q$ (q is the total output)
Cost: each competitive firm has cost function C(q) = 200q

Competitive equilibrium

- Total Output: $q = 600$
- Price: 200
- Total profit: 0

Cournot Duopoly

- Output: $q_1 = q_2 = 200$
- Price $p(q_1 + q_2) = 300$
- Each firm's profit

$\Pi_1 = 20000.$

$\Pi_2 = 20000.$

Compare a duopoly with competitive equilibrium (summary)

- Hence, a Cournot equilibrium has lower (total) output level, higher price, and higher (total) profit than a competitive market.